

Analyzing E-Learning Platform Purchases using MySQL

Project Overview

This project involved designing and analyzing an e-learning platform database using MySQL. Three tables—learners, courses, and purchases—were created with primary and foreign key relationships. SQL queries were used to analyze learner purchases, course performance, and sales trends.

Key Tasks Performed

- Created a relational database and inserted sample data.
- Used INNER JOIN, LEFT JOIN, and RIGHT JOIN to combine data.
- Applied aggregate functions to calculate total spending, revenue, and purchase counts.
- Used subqueries, CTE, CASE expression, NULL handling, and VIEW for advanced SQL analysis.
- Created category_performance_view to summarize category-wise performance.

Key Insights

- Identified learners with the highest spending.
- Found the top-performing courses based on purchase quantity.
- Analyzed revenue generated by each course category.
- Identified courses that were never purchased.
- Demonstrated handling of NULL values to improve data quality.

Recommendations

- Focus marketing on high-performing course categories.
- Promote less popular courses using discounts or bundles.
- Regularly clean and validate missing data for accurate reporting.
- Use SQL views to simplify reporting and improve decision-making.

Conclusion

This project enhanced my practical knowledge of MySQL by applying joins, aggregate functions, subqueries, CTEs, CASE expressions, NULL handling, and views to generate meaningful business insights from e-learning purchase data.